

# Лабораторная работа №2

Простые сети в GNS3. Анализ трафика

---

Гашимова Эсма Эльшан кызы

2026-09-20

1. Цели и задачи работы
2. Сеть на базе Ethernet-коммутатора
3. Анализ трафика в Wireshark
4. Сеть на базе маршрутизатора FRR
5. Сеть на базе маршрутизатора VyOS
6. Результаты работы

## 1. 1. Цели и задачи работы

---

## 1.1 Цель лабораторной работы

Получение практических навыков моделирования простых компьютерных сетей в GNS3.

Изучение настройки IP-адресации узлов и маршрутизаторов.

Исследование сетевого трафика с использованием Wireshark.

## 2. 2. Сеть на базе Ethernet-коммутатора

---

## 2.1 Создание топологии

В GNS3 создана сеть, состоящая из двух оконечных устройств VPCS и Ethernet-коммутатора.

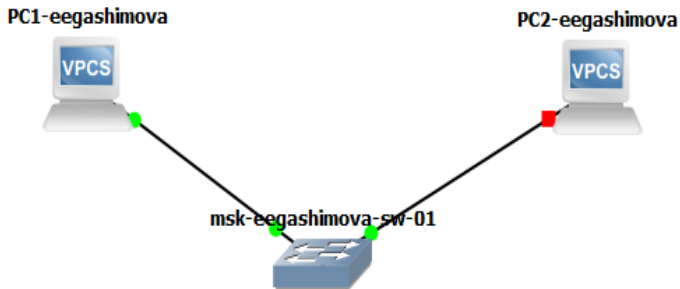
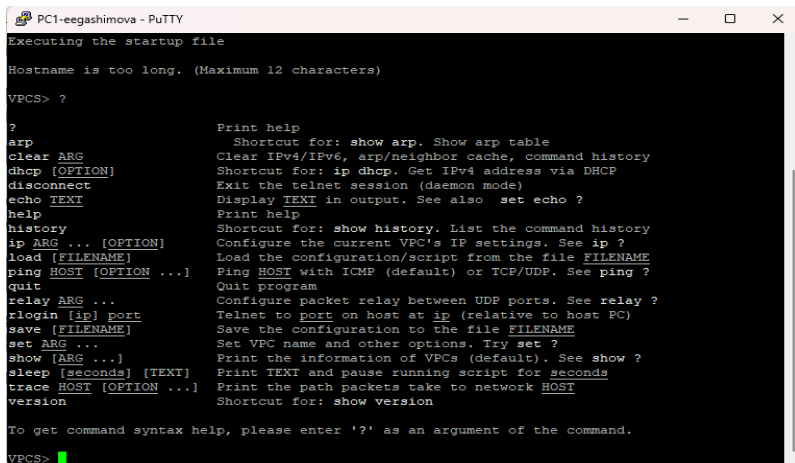


Рисунок 1: Топология простейшей сети на базе Ethernet-коммутатора

## 2.2 Команды VPCS

Для настройки и диагностики оконечных устройств использовался интерфейс командной строки VPCS.



```
PC1-eegashimova - PuTTY
Executing the startup file
Hostname is too long. (Maximum 12 characters)
VPCS> ?

?                Print help
arp              Shortcut for: show arp. Show arp table
clear ARG        Clear IPv4/IPv6, arp/neighbor cache, command history
dhcp [OPTION]    Shortcut for: ip dhcp. Get IPv4 address via DHCP
disconnect       Exit the telnet session (daemon mode)
echo TEXT        Display TEXT in output. See also set echo ?
help             Print help
history          Shortcut for: show history. List the command history
ip ARG ... [OPTION] Configure the current VPC's IP settings. See ip ?
load [FILENAME]  Load the configuration/script from the file FILENAME
ping HOST [OPTION ...] Ping HOST with ICMP (default) or TCP/UDP. See ping ?
quit            Quit program
relay ARG ...    Configure packet relay between UDP ports. See relay ?
rlogin [ip] port Telnet to port on host at ip (relative to host PC)
save [FILENAME]  Save the configuration to the file FILENAME
set ARG ...      Set VPC name and other options. Try set ?
show [ARG ...]   Print the information of VPCs (default). See show ?
sleep [seconds] [TEXT] Print TEXT and pause running script for seconds
trace HOST [OPTION ...] Print the path packets take to network HOST
version          Shortcut for: show version

To get command syntax help, please enter '?' as an argument of the command.
VPCS>
```

## 2.3 Настройка PC1

Узлу **PC1-eegashimova** назначен IP-адрес **192.168.1.11/24**.

В качестве шлюза указан адрес **192.168.1.1**.

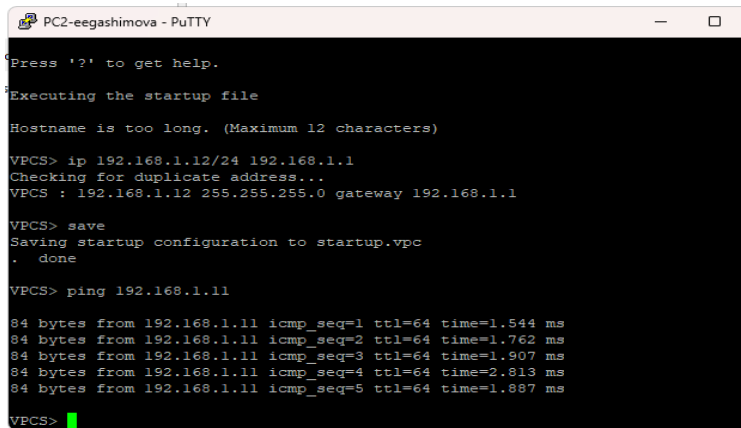


```
VPCS>  
VPCS> ip 192.168.1.11/24 192.168.1.1  
Checking for duplicate address...  
VPCS : 192.168.1.11 255.255.255.0 gateway 192.168.1.1  
  
VPCS> save  
Saving startup configuration to startup.vpc  
. done  
  
VPCS>
```

## 2.4 Настройка PC2 и проверка связи

Узлу **PC2-eegashimova** назначен IP-адрес **192.168.1.12/24**.

Связь между узлами проверена командой **ping**.



```
PC2-eegashimova - PuTTY
Press '?' to get help.
Executing the startup file
Hostname is too long. (Maximum 12 characters)

VPCS> ip 192.168.1.12/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> ping 192.168.1.11

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=1.544 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=1.762 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=1.907 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=2.813 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=1.887 ms

VPCS>
```

### 3. 3. Анализ трафика в Wireshark

---

## 3.1 Захват сетевого трафика

На соединении между **PC1-eegashimova** и Ethernet-коммутатором включён захват пакетов.

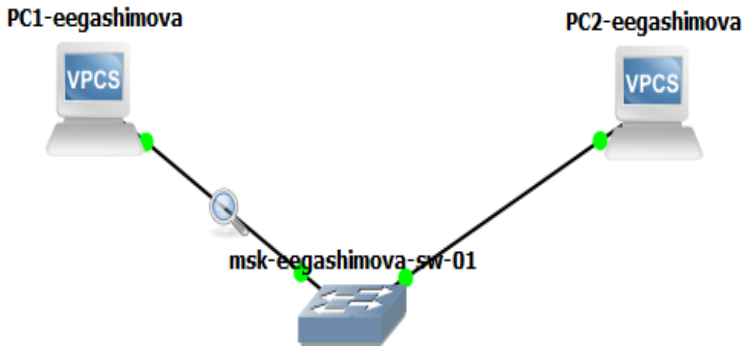


Рисунок 5: Захват трафика на соединении между PC1 и коммутатором

### 3.2 Анализ ARP

В Wireshark зафиксированы ARP-сообщения.

Gratuitous ARP используется узлом для объявления собственного IP-адреса и обнаружения возможного конфликта адресов.

| No. | Time     | Source           | Destination | Protocol | Length | Info                                      |
|-----|----------|------------------|-------------|----------|--------|-------------------------------------------|
| 1   | 0.000000 | ::               | ff02::2     | ICMPv6   | 62     | Router Solicitation                       |
| 2   | 0.002486 | ::               | ff02::2     | ICMPv6   | 62     | Router Solicitation                       |
| 3   | 0.050523 | Private_66:68:00 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.11 (Request) |
| 4   | 0.053474 | Private_66:68:01 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.12 (Request) |
| 5   | 1.051904 | Private_66:68:00 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.11 (Request) |
| 6   | 1.053876 | Private_66:68:01 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.12 (Request) |
| 7   | 2.052127 | Private_66:68:00 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.11 (Request) |
| 8   | 2.054894 | Private_66:68:01 | Broadcast   | ARP      | 64     | Gratuitous ARP for 192.168.1.12 (Request) |

```

    Frame 6: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
    Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
    > Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    > Source: Private_66:68:01 (00:50:79:66:68:01)
    Type: ARP (0x0806)
    [Stream index: 3]
    Padding: 00000000000000000000000000000000
    Frame check sequence: 0x00000000 [unverified]
    [FCS Status: Unverified]
    Address Resolution Protocol (request/gratuitous ARP)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    [Is gratuitous: True]
    Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
    Sender IP address: 192.168.1.12
    Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
    Target IP address: 192.168.1.12

```

```
0000 ff ff ff ff ff ff 00 50 79 66 68 01 05 06 00 01  ...
0010 08 00 06 04 00 01 00 50 79 66 68 01 c0 a8 01 0c  ...
0020 ff ff ff ff ff ff c0 a8 01 0c 00 00 00 00 00 00  ...
0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ...
```

## 3.3 Проверка различных режимов ping

В VPCS выполнена проверка соединения в трёх режимах:

- ICMP;

Во всех случаях удалённый узел успешно отвечает.

```
PC2-eegashimova - PuTTY
VPCS : 192.168.1.12 255.255.255.0 gateway 192.168.1.1
VPCS> ping ?

ping HOST [OPTION ...]
Ping the network HOST. HOST can be an ip address or name
Options:
  -i          ICMP mode, default
  -2          UDP mode
  -3          TCP mode
  -c count    Packet count, default 5
  -D          Set the Don't Fragment bit
  -f FLAG     Tcp header FLAG |C|E|U|A|P|R|S|F|
               bits |7 6 5 4 3 2 1 0|
  -i ms       Wait ms milliseconds between sending each packet
  -l size     Data Size
  -P protocol Use IP protocol in ping packets
               1 - ICMP (default), 17 - UDP, 6 - TCP
  -P port     Destination port
  -S port     Source port
  -T ttl      Set ttl, default 64
  -t          Send packets until interrupted by Ctrl+C
  -w ms       Wait ms milliseconds to receive the response

Notes: 1. Using names requires DNS to be set.
       2. Use Ctrl+C to stop the command.

VPCS>
VPCS> ping 192.168.1.11 -c 1 -i
84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=2.009 ms
VPCS> ping 192.168.1.11 -c 1 -2
84 bytes from 192.168.1.11 udp_seq=1 ttl=64 time=2.009 ms
VPCS> ping 192.168.1.11 -c 1 -3
84 bytes from 192.168.1.11 tcp_seq=1 ttl=64 time=2.009 ms
```

### 3.3 Проверка различных режимов ping

В VPCS выполнена проверка соединения в трёх режимах:

- ICMP;
- UDP;

Во всех случаях удалённый узел успешно отвечает.

```
PC2-eegashimova - PuTTY
VPCS : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

VPCS> ping ?

ping HOST [OPTION ...]
Ping the network HOST. HOST can be an ip address or name
Options:
  -i          ICMP mode, default
  -2          UDP mode
  -3          TCP mode
  -c count    Packet count, default 5
  -D          Set the Don't Fragment bit
  -f FLAG     Tcp header FLAG |C|E|U|A|P|R|S|F|
               bits |7 6 5 4 3 2 1 0|
  -i ms       Wait ms milliseconds between sending each packet
  -l size     Data Size
  -P protocol  Use IP protocol in ping packets
               1 - ICMP (default), 17 - UDP, 6 - TCP
  -p port     Destination port
  -s port     Source port
  -T ttl      Set ttl, default 64
  -t          Send packets until interrupted by Ctrl+C
  -w ms       Wait ms milliseconds to receive the response

Notes: 1. Using names requires DNS to be set.
       2. Use Ctrl+C to stop the command.

VPCS>
VPCS> ping 192.168.1.11 -c 1 -1

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=2.009 ms

VPCS> ping 192.168.1.11 -c 1 -2
```

## 3.3 Проверка различных режимов ping

В VPCS выполнена проверка соединения в трёх режимах:

- ICMP;
- UDP;
- TCP.

Во всех случаях удалённый узел успешно отвечает.

```
PC2-eegashimova - PuTTY
VPCS : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

VPCS> ping ?

ping HOST [OPTION ...]
Ping the network HOST. HOST can be an ip address or name
Options:
  -i          ICMP mode, default
  -2          UDP mode
  -3          TCP mode
  -c count    Packet count, default 5
  -D          Set the Don't Fragment bit
  -f FLAG     Tcp header FLAG |C|E|U|A|P|R|S|F|
               bits |7 6 5 4 3 2 1 0|
  -i ms       Wait ms milliseconds between sending each packet
  -l size     Data size
  -P protocol 1 - ICMP (default), 17 - UDP, 6 - TCP
  -P port     Destination port
  -S port     Source port
  -T ttl       Set ttl, default 64
  -t          Send packets until interrupted by Ctrl+C
  -w ms       Wait ms milliseconds to receive the response

Notes: 1. Using names requires DNS to be set.
       2. Use Ctrl+C to stop the command.

VPCS>
VPCS> ping 192.168.1.11 -c 1 -1

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=2.009 ms

VPCS> ping 192.168.1.11 -c 1 -2
```

## 3.4 Анализ ICMP

При стандартной проверке **ping** передаются сообщения **ICMP Echo Request** и **ICMP Echo Reply**.

Наличие ответного пакета подтверждает доступность удалённого узла.

| No. | Time       | Source           | Destination      | Protocol | Length | Info                                                                       |
|-----|------------|------------------|------------------|----------|--------|----------------------------------------------------------------------------|
| 8   | 2.094804   | Private_66:68:01 | Broadcast        | ARP      | 64     | Gratuitous ARP for 192.168.1.12 (Request)                                  |
| 9   | 225.437670 | Private_66:68:01 | Broadcast        | ARP      | 64     | who has 192.168.1.11? Tell 192.168.1.12                                    |
| 10  | 225.439669 | Private_66:68:00 | Private_66:68:01 | ARP      | 64     | 192.168.1.11 is at 00:50:79:66:68:00                                       |
| 11  | 225.441485 | 192.168.1.12     | 192.168.1.11     | ICMP     | 98     | Echo (ping) request id=0x3ad4, seq=1/256, ttl=64 (reply in 12)             |
| 12  | 225.442257 | 192.168.1.11     | 192.168.1.12     | ICMP     | 98     | Echo (ping) reply id=0x3ad4, seq=1/256, ttl=64 (request in 11)             |
| 13  | 229.636874 | 192.168.1.12     | 192.168.1.11     | ECHO     | 98     | Request                                                                    |
| 14  | 229.637993 | 192.168.1.11     | 192.168.1.12     | ECHO     | 98     | Response                                                                   |
| 15  | 232.253722 | 192.168.1.12     | 192.168.1.11     | TCP      | 74     | 6302 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=1789908033 TSecr=0 WS=2 |
| 16  | 232.254922 | 192.168.1.11     | 192.168.1.12     | TCP      | 54     | 7 → 6302 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0                             |
| 17  | 232.257077 | 192.168.1.12     | 192.168.1.11     | TCP      | 66     | 6302 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1789908033 TSecr=0         |
| 18  | 232.258135 | 192.168.1.12     | 192.168.1.11     | ECHO     | 122    | Request                                                                    |

|                                                                                                     |                                                                   |
|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| > Frame 11: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0  | 0000 00 50 79 66 68 00 00 50 79 66 68 01 08 00 45 00 ·Pyfh·P yf   |
| > Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00) | 0010 00 54 d4 3a 00 00 40 01 23 07 c0 a8 01 0c c0 a8 ·T.:. @. #   |
| > Destination: Private_66:68:00 (00:50:79:66:68:00)                                                 | 0020 01 00 08 00 c5 36 3a d4 00 01 08 09 0a 00 0c 0d ·····G: ··   |
| > Source: Private_66:68:01 (00:50:79:66:68:01)                                                      | 0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d ······       |
| Type: IPv4 (0x0800)                                                                                 | 0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d ···!#\$% &   |
| [Stream index: 4]                                                                                   | 0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d ·./012345 67 |
| > Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11                                 | 0060 3e 3f ······                                                 |
| 0100 ···· = Version: 4                                                                              |                                                                   |
| ···· 0101 = Header Length: 20 bytes (5)                                                             |                                                                   |
| > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)                                     |                                                                   |
| Total Length: 84                                                                                    |                                                                   |
| Identification: 0xd43a (54330)                                                                      |                                                                   |
| > 000 ···· = Flags: 0x0                                                                             |                                                                   |
| ···· 0000 0000 0000 = Fragment Offset: 0                                                            |                                                                   |
| Time to Live: 64                                                                                    |                                                                   |
| Protocol: ICMP (1)                                                                                  |                                                                   |
| Header Checksum: 0x2307 [validation disabled]                                                       |                                                                   |
| [Header checksum status: Unverified]                                                                |                                                                   |
| Source Address: 192.168.1.12                                                                        |                                                                   |
| Destination Address: 192.168.1.11                                                                   |                                                                   |
| [Stream index: 0]                                                                                   |                                                                   |
| > Internet Control Message Protocol                                                                 |                                                                   |
| Type: Echo (ping) request (8)                                                                       |                                                                   |
| Code: 0                                                                                             |                                                                   |
| Checksum: 0xe536 [correct]                                                                          |                                                                   |
| [Checksum Status: Good]                                                                             |                                                                   |
| Identifier (BE): 15060 (0x3ad4)                                                                     |                                                                   |
| Identifier (LE): 54330 (0xd43a)                                                                     |                                                                   |
| Sequence Number (BE): 1 (0x0001)                                                                    |                                                                   |
| Sequence Number (LE): 256 (0x0100)                                                                  |                                                                   |
| [Response frame: 12]                                                                                |                                                                   |
| > Data (56 bytes)                                                                                   |                                                                   |

## 3.5 Анализ UDP

При проверке в UDP-режиме между узлами передаются UDP-дейтаграммы.

UDP не требует предварительного установления соединения между устройствами.

| No. | Time       | Source           | Destination      | Protocol | Length | Info                                                                       |
|-----|------------|------------------|------------------|----------|--------|----------------------------------------------------------------------------|
| 8   | 2.0548804  | Private_66:68:01 | Broadcast        | ARP      | 64     | Gratuitous ARP for 192.168.1.12 (Request)                                  |
| 9   | 225.437670 | Private_66:68:01 | Broadcast        | ARP      | 64     | Who has 192.168.1.11? Tell 192.168.1.12                                    |
| 10  | 225.439669 | Private_66:68:00 | Private_66:68:01 | ARP      | 64     | 192.168.1.11 is at 00:50:79:66:68:00                                       |
| 11  | 225.441485 | 192.168.1.12     | 192.168.1.11     | ICMP     | 98     | Echo (ping) request id=0x3ad4, seq=1/256, ttl=64 (reply in 12)             |
| 12  | 225.442257 | 192.168.1.11     | 192.168.1.12     | ICMP     | 98     | Echo (ping) reply id=0x3ad4, seq=1/256, ttl=64 (request in 11)             |
| 13  | 229.636874 | 192.168.1.12     | 192.168.1.11     | ECHO     | 98     | Request                                                                    |
| 14  | 229.637993 | 192.168.1.11     | 192.168.1.12     | ECHO     | 98     | Response                                                                   |
| 15  | 232.253272 | 192.168.1.12     | 192.168.1.11     | TCP      | 74     | 6302 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=1789908033 TSecr=0 WS=2 |
| 16  | 232.254922 | 192.168.1.11     | 192.168.1.12     | TCP      | 54     | 7 → 6302 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0                             |
| 17  | 232.257077 | 192.168.1.12     | 192.168.1.11     | TCP      | 66     | 6302 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1789908033 TSecr=0         |
| 18  | 232.258135 | 192.168.1.12     | 192.168.1.11     | ECHO     | 122    | Request                                                                    |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>&gt; Frame 14: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0</div> <div>Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:01 (00:50:79:66:68:01)</div> <div>&gt; Destination: Private_66:68:01 (00:50:79:66:68:01)</div> <div>&gt; Source: Private_66:68:00 (00:50:79:66:68:00)</div> <div>Type: IPv4 (0x0800)</div> <div>[Stream index: 4]</div> <div>Internet Protocol Version 4, Src: 192.168.1.12, Dst: 192.168.1.11</div> <div>0100 .... = Version: 4</div> <div>.... 0101 = Header Length: 20 bytes (5)</div> <div>&gt; Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)</div> <div>Total Length: 84</div> <div>Identification: 0xd43e (54334)</div> <div>&gt; 0000 .... = Flags: 0x0</div> <div>...0 0000 0000 0000 = Fragment Offset: 0</div> <div>Time to Live: 64</div> <div>Protocol: UDP (17)</div> <div>Header Checksum: 0x22f3 [validation disabled]</div> <div>[Header checksum status: Unverified]</div> <div>Source Address: 192.168.1.12</div> <div>Destination Address: 192.168.1.11</div> <div>[Stream index: 0]</div> <div>User Datagram Protocol, Src Port: 7, Dst Port: 55530</div> <div>Source Port: 7</div> <div>Destination Port: 55530</div> <div>Length: 64</div> <div>Checksum: 0x078a [unverified]</div> <div>[Checksum Status: Unverified]</div> <div>[Stream index: 0]</div> <div>[Stream Packet Number: 2]</div> <div>&gt; [Timestamps]</div> <div>UDP payload (56 bytes)</div> <div>Echo</div> <div>Echo data: 005079666801000f101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f3031323334353637...</div> | <div>0000 00 50 79 66 68 01 00 50 79 66 68 00 08 00 45 00 ..Pyfh..P..yf</div> <div>0010 00 54 d4 3e 00 00 40 11 22 f3 c0 a8 01 0b c0 a8 ..T&gt;..@..</div> <div>0020 01 0c 00 07 d5 ea 00 40 07 5a 00 50 79 66 68 01 .....@..</div> <div>0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d ..... ..</div> <div>0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d .....!"#\$%&amp;'</div> <div>0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d .../012345 67</div> <div>0060 3e 3f &gt;</div> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 3.6 Анализ TCP

При проверке в TCP-режиме перед передачей данных выполняется установка соединения:

**SYN → SYN, ACK → ACK.**

После передачи данных соединение завершается с использованием флагов **FIN** и **ACK**.

| No. | Time       | Source       | Destination  | Protocol | Length | Info                                                                          |
|-----|------------|--------------|--------------|----------|--------|-------------------------------------------------------------------------------|
| 13  | 229.636874 | 192.168.1.12 | 192.168.1.11 | ECHO     | 98     | Request                                                                       |
| 14  | 229.637993 | 192.168.1.11 | 192.168.1.12 | ECHO     | 98     | Response                                                                      |
| 15  | 232.253272 | 192.168.1.12 | 192.168.1.11 | TCP      | 74     | 6302 → 7 [SYN] Seq=0 Win=2920 Len=0 MSS=1460 TSval=1789908033 TSecr=0 WS=2    |
| 16  | 232.254922 | 192.168.1.11 | 192.168.1.12 | TCP      | 54     | 7 → 6302 [SYN, ACK] Seq=0 Ack=1 Win=2920 Len=0                                |
| 17  | 232.257077 | 192.168.1.12 | 192.168.1.11 | TCP      | 66     | 6302 → 7 [ACK] Seq=1 Ack=1 Win=2920 Len=0 TSval=1789908033 TSecr=0            |
| 18  | 232.258135 | 192.168.1.12 | 192.168.1.11 | ECHO     | 122    | Request                                                                       |
| 19  | 232.259463 | 192.168.1.11 | 192.168.1.12 | TCP      | 54     | 7 → 6302 [ACK] Seq=1 Ack=57 Win=2920 Len=0                                    |
| 20  | 232.262929 | 192.168.1.12 | 192.168.1.11 | TCP      | 66     | 6302 → 7 [FIN, PSH, ACK] Seq=57 Ack=1 Win=2920 Len=0 TSval=1789908033 TSecr=0 |
| 21  | 232.263332 | 192.168.1.11 | 192.168.1.12 | TCP      | 54     | 7 → 6302 [ACK] Seq=1 Ack=58 Win=2920 Len=0                                    |
| 22  | 232.263371 | 192.168.1.11 | 192.168.1.12 | TCP      | 54     | 7 → 6302 [FIN, ACK] Seq=1 Ack=58 Win=2920 Len=0                               |
| 23  | 232.265841 | 192.168.1.12 | 192.168.1.11 | TCP      | 66     | 6302 → 7 [ACK] Seq=58 Ack=2 Win=2920 Len=0 TSval=1789908033 TSecr=0           |

Time to Live: 64  
Protocol: TCP (6)  
Header checksum: 0x22e1 [validation disabled]  
[Header checksum status: Unverified]  
Source Address: 192.168.1.12  
Destination Address: 192.168.1.11  
[Stream index: 0]  
Transmission Control Protocol, Src Port: 6302, Dst Port: 7, Seq: 1, Ack: 1, Len: 56  
Source Port: 6302  
Destination Port: 7  
[Stream index: 0]  
[Stream Packet Number: 4]  
> [Conversation completeness: Complete, WITH\_DATA (31)]  
[TCP Segment Len: 56]  
Sequence Number: 1 (relative sequence number)  
Sequence Number (raw): 531755281  
[Next Sequence Number: 57 (relative sequence number)]  
Acknowledgment Number: 1 (relative ack number)  
Acknowledgment number (raw): 531755281  
1600 ... = Header Length: 32 bytes (8)  
> Flags: 0x018 (PSH, ACK)  
Window: 2920  
[Calculated window size: 2920]  
[Window size scaling factor: -2 (no window scaling used)]  
Checksum: 0x0178 [unverified]  
[Checksum Status: Unverified]  
Urgent Pointer: 0  
> Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps  
> [Timestamps]  
> [SEQ/ACK analysis]  
> [Client Contiguous Streams: 1]  
> [Server Contiguous Streams: 1]  
TCP payload (56 bytes)  
Echo data: 0000 00 50 79 66 68 00 00 50 79 66 68 01 08 00 45 00 -Pyfh-P yf  
0010 00 6c d4 43 00 00 40 06 22 e1 c0 a8 01 0c c0 a8 -1C-@-  
0020 01 0b 15 9a 00 07 1f b1 f1 11 1f b1 f1 11 00 15 -h-x-  
0030 0b 68 01 78 00 00 01 01 08 0a 6a af d4 41 00 00 -h-x-  
0040 00 00 0d 0a 0d 0a 0d 0a 0d 0a 0d 0a 0d 0a 0d 0a -  
0050 0d 0a 0d 0a 0d 0a 0d 0a 0d 0a 0d 0a 0d 0a 0d 0a -  
0060 0d 0a 0d 0a 0d 0a 0d 0a 0d 0a 0d 0a 0d 0a 0d 0a -  
0070 0d 0a 0d 0a 0d 0a 0d 0a 0d 0a 0d 0a 0d 0a 0d 0a -

## 4. 4. Сеть на базе маршрутизатора FRR

---

## 4.1 Топология с FRR

В GNS3 создана сеть, состоящая из VPCS, Ethernet-коммутатора и маршрутизатора FRR.

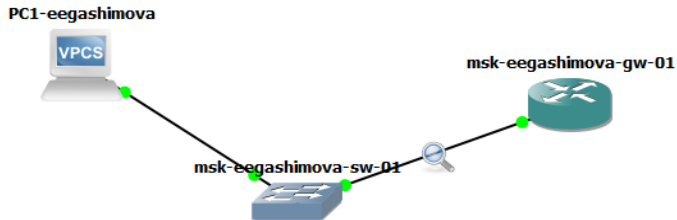


Рисунок 11: Топология сети с маршрутизатором FRR

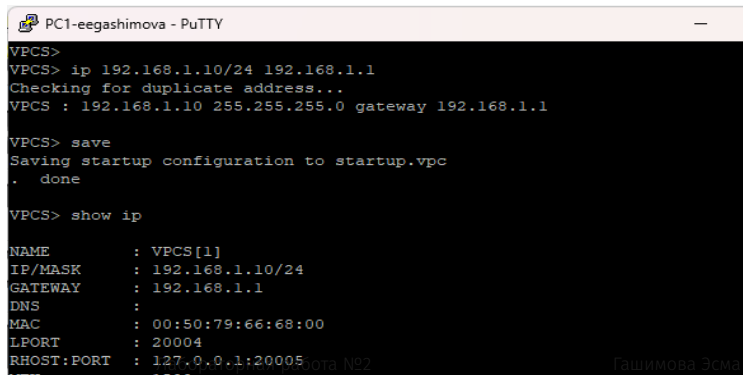
## 4.2 Настройка оконечного устройства

Узлу **PC1-eegashimova** назначен адрес:

**192.168.1.10/24**

В качестве шлюза указан маршрутизатор:

**192.168.1.1**



```
PC1-eegashimova - PuTTY
VPCS>
VPCS> ip 192.168.1.10/24 192.168.1.1
Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip
NAME           : VPCS[1]
IP/MASK        : 192.168.1.10/24
GATEWAY        : 192.168.1.1
DNS            :
MAC            : 00:50:79:66:68:00
LPORT          : 20004
RHOST:PORT     : 127.0.0.1:20005
RHOST
```

## 4.3 Настройка маршрутизатора FRR

Маршрутизатору присвоено имя **msk-eegashimova-gw-01**.

Интерфейсу **eth0** назначен адрес **192.168.1.1/24**.

```
frr# configure terminal
frr(config)# hostname msk-eegashimova-gw-01
msk-eegashimova-gw-01(config)# exit
msk-eegashimova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-eegashimova-gw-01# configure terminal
msk-eegashimova-gw-01(config)# interface eth0
msk-eegashimova-gw-01(config-if)# ip address 192.168.1.1/24
msk-eegashimova-gw-01(config-if)# no shutdown
msk-eegashimova-gw-01(config-if)# exit
msk-eegashimova-gw-01(config)# exit
msk-eegashimova-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-eegashimova-gw-01#
```

## 4.4 Проверка конфигурации FRR

Для проверки настроек использованы команды:

**show running-config**

**show interface brief**

Интерфейс **eth0** находится в состоянии **up**.

```
msk-eegashimova-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-eegashimova-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 192.168.1.1/24
exit
!
end
msk-eegashimova-gw-01# show interface brief
Interface      Status    VRF        Addresses
-----
eth0            up        default    192.168.1.1/24
eth1            down      default
eth2            down      default
eth3            down      default
eth4            down      default
eth5            down      default
eth6            down      default
eth7            down      default
```

## 4.5 Проверка связи с FRR

С узла **PC1-eegashimova** выполнена команда:

**ping 192.168.1.1**

Получены ответы от маршрутизатора без потери пакетов.

```
VPCS>  
VPCS> ping 192.168.1.1  
  
64 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=2.936 ms  
64 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=3.378 ms  
64 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=3.342 ms  
64 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=2.297 ms  
64 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.668 ms
```

## 4.6 Анализ трафика FRR

Перед передачей ICMP-пакетов выполняется ARP-разрешение IP-адреса маршрутизатора.

После этого между **192.168.1.10** и **192.168.1.1** передаются сообщения **Echo Request** и **Echo Reply**.

| No. | Time       | Source            | Destination      | Protocol | Length | Info                                                            |
|-----|------------|-------------------|------------------|----------|--------|-----------------------------------------------------------------|
| 16  | 179.162435 | 0c:df:47:6d:00:00 | Private:66:68:00 | ARP      | 60     | 192.168.1.1 is at 0c:df:47:6d:00:00                             |
| 17  | 179.164694 | 192.168.1.10      | 192.168.1.1      | ICMP     | 98     | Echo (ping) request id=0xbad5, seq=1/256, ttl=64 (reply in 18)  |
| 18  | 179.166981 | 192.168.1.1       | 192.168.1.10     | ICMP     | 98     | Echo (ping) reply id=0xbad5, seq=1/256, ttl=64 (request in 17)  |
| 19  | 180.167879 | 192.168.1.10      | 192.168.1.1      | ICMP     | 98     | Echo (ping) request id=0xbbd5, seq=2/512, ttl=64 (reply in 20)  |
| 20  | 180.170182 | 192.168.1.1       | 192.168.1.10     | ICMP     | 98     | Echo (ping) reply id=0xbbd5, seq=2/512, ttl=64 (request in 19)  |
| 21  | 181.173483 | 192.168.1.10      | 192.168.1.1      | ICMP     | 98     | Echo (ping) request id=0xbcd5, seq=3/768, ttl=64 (reply in 22)  |
| 22  | 181.175327 | 192.168.1.1       | 192.168.1.10     | ICMP     | 98     | Echo (ping) reply id=0xbcd5, seq=3/768, ttl=64 (request in 21)  |
| 23  | 182.176946 | 192.168.1.10      | 192.168.1.1      | ICMP     | 98     | Echo (ping) request id=0xbdd5, seq=4/1024, ttl=64 (reply in 24) |
| 24  | 182.178397 | 192.168.1.1       | 192.168.1.10     | ICMP     | 98     | Echo (ping) reply id=0xbdd5, seq=4/1024, ttl=64 (request in 23) |
| 25  | 183.180939 | 192.168.1.10      | 192.168.1.1      | ICMP     | 98     | Echo (ping) request id=0xbed5, seq=5/1280, ttl=64 (reply in 26) |

|                                                                                                           |                                                                  |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Character encoding: ASCII (0)                                                                             | 0000 0c df 47 6d 00 00 00 50 79 66 68 00 08 00 45 00 ..Gm...P yf |
| [Coloring Rule Name: ICMP]                                                                                | 0010 00 54 d5 bb 00 00 40 01 21 92 c0 a8 01 0a c0 a8 .T...@. !:  |
| [Coloring Rule String: icmpv6]                                                                            | 0020 01 01 08 00 64 34 bb 05 00 02 08 09 0a 00 0c 0d ...-d4- ..  |
| Ethernet II, Src: Private:66:68:00 (00:50:79:66:68:00), Dst: 0c:df:47:6d:00:00 (0c:df:47:6d:00:00)        | 0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d .....-..    |
| > Destination: 0c:df:47:6d:00:00 (0c:df:47:6d:00:00)                                                      | 0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d ...!##% &'  |
| > Source: Private:66:68:00 (00:50:79:66:68:00)                                                            | 0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d ./012345 67 |
| Type: IPv4 (0x0800)                                                                                       | 0060 3e 3f ..?                                                   |
| [Stream index: 4]                                                                                         |                                                                  |
| Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.1                                          |                                                                  |
| 0100 .... = Version: 4                                                                                    |                                                                  |
| .... 0101 = Header Length: 20 bytes (5)                                                                   |                                                                  |
| > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)                                           |                                                                  |
| Total Length: 84                                                                                          |                                                                  |
| Identification: 0xd5bb (54715)                                                                            |                                                                  |
| > 0000 .... = Flags: 0x0                                                                                  |                                                                  |
| ...0 0000 0000 0000 = Fragment Offset: 0                                                                  |                                                                  |
| Time to Live: 64                                                                                          |                                                                  |
| Protocol: ICMP (1)                                                                                        |                                                                  |
| Header Checksum: 0x2192 [validation disabled]                                                             |                                                                  |
| [Header checksum status: Unverified]                                                                      |                                                                  |
| Source Address: 192.168.1.10                                                                              |                                                                  |
| Destination Address: 192.168.1.1                                                                          |                                                                  |
| [Stream index: 0]                                                                                         |                                                                  |
| Internet Control Message Protocol                                                                         |                                                                  |
| Type: Echo (ping) request (8)                                                                             |                                                                  |
| Code: 0                                                                                                   |                                                                  |
| Checksum: 0x6434 [correct]                                                                                |                                                                  |
| [Checksum Status: Good]                                                                                   |                                                                  |
| Identifier (BE): 48095 (0xbbd5)                                                                           |                                                                  |
| Identifier (LE): 54715 (0xd5bb)                                                                           |                                                                  |
| Sequence Number (BE): 2 (0x0002)                                                                          |                                                                  |
| Sequence Number (LE): 512 (0x0200)                                                                        |                                                                  |
| [Response frame: 20]                                                                                      |                                                                  |
| Data (56 bytes)                                                                                           |                                                                  |
| Data: 00000000c00000ff101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f3031323334353637... |                                                                  |
| [Length: 56]                                                                                              |                                                                  |

## 5. 5. Сеть на базе маршрутизатора VyOS

---

## 5.1 Топология с VyOS

Создана сеть, состоящая из VPCS, Ethernet-коммутатора и маршрутизатора VyOS.

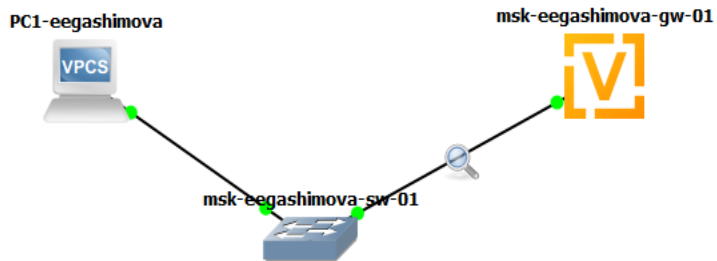
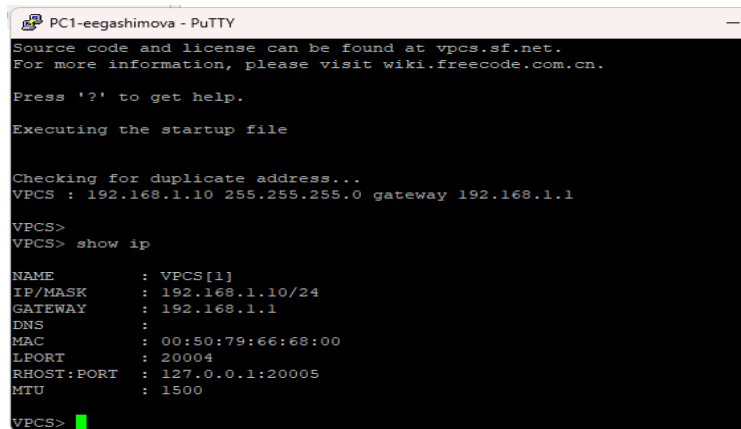


Рисунок 17: Топология сети с маршрутизатором VyOS

## 5.2 Настройка оконечного устройства

Узлу **PC1-eegashimova** назначен адрес **192.168.1.10/24**.

Шлюзом по умолчанию является адрес **192.168.1.1**.



```
PC1-eegashimova - PuTTY
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

Checking for duplicate address...
VPCS : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

VPCS>
VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.1.10/24
GATEWAY    : 192.168.1.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 20004
RHOST:PORT : 127.0.0.1:20005
MTU        : 1500

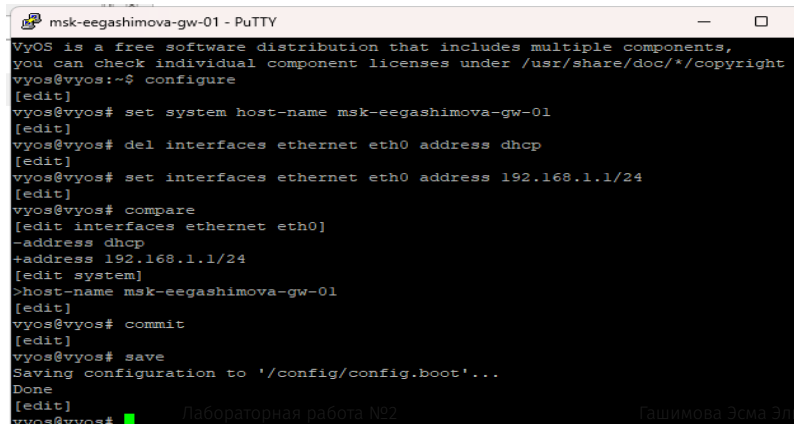
VPCS>
```

## 5.3 Настройка маршрутизатора VyOS

Для интерфейса **eth0** удалено получение IP-адреса по DHCP.

Интерфейсу назначен статический адрес **192.168.1.1/24**.

Конфигурация применена и сохранена.



```
msk-eegashimova-gw-01 - PuTTY
VyOS is a free software distribution that includes multiple components,
you can check individual component licenses under /usr/share/doc/*/copyright
vyos@vyos:~$ configure
[edit]
vyos@vyos# set system host-name msk-eegashimova-gw-01
[edit]
vyos@vyos# del interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# set interfaces ethernet eth0 address 192.168.1.1/24
[edit]
vyos@vyos# compare
[edit interfaces ethernet eth0]
-address dhcp
+address 192.168.1.1/24
[edit system]
>host-name msk-eegashimova-gw-01
[edit]
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
vyos@vyos#
```

## 5.4 Проверка интерфейсов VyOS

Командой **show interfaces** проверено состояние сетевых интерфейсов маршрутизатора.

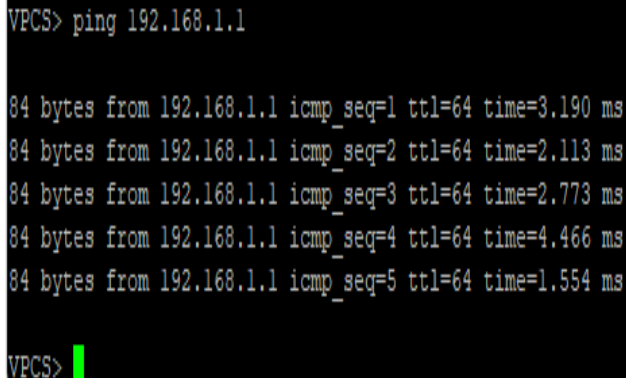
Интерфейс **eth0** имеет адрес **192.168.1.1/24**.

```
[edit]
vyos@vyos# show interfaces
  ethernet eth0 {
    address 192.168.1.1/24
    hw-id 0c:b2:f7:47:00:00
  }
  ethernet eth1 {
    hw-id 0c:b2:f7:47:00:01
  }
  ethernet eth2 {
    hw-id 0c:b2:f7:47:00:02
  }
  loopback lo {
  }
[edit]
vyos@vyos#
```

## 5.5 Проверка связи с VyOS

С узла **PC1-eegashimova** выполнена проверка доступности маршрутизатора.

Все ICMP-запросы получили ответы.



```
VPCS> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=3.190 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=2.113 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=2.773 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=4.466 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.554 ms

VPCS> 
```

## 5.6 Анализ трафика VyOS

В Wireshark зафиксирован ARP-обмен между компьютером и маршрутизатором.

После определения MAC-адресов выполняется обмен сообщениями **ICMP Echo Request** и **ICMP Echo Reply**.

| No. | Time       | Source            | Destination       | Protocol | Length | Info                                                            |
|-----|------------|-------------------|-------------------|----------|--------|-----------------------------------------------------------------|
| 23  | 161.625132 | 192.168.1.10      | 192.168.1.1       | ICMP     | 98     | Echo (ping) request id=0x4cd7, seq=2/512, ttl=64 (reply in 24)  |
| 24  | 161.626504 | 192.168.1.1       | 192.168.1.10      | ICMP     | 98     | Echo (ping) reply id=0x4cd7, seq=2/512, ttl=64 (request in 23)  |
| 25  | 162.627770 | 192.168.1.10      | 192.168.1.1       | ICMP     | 98     | Echo (ping) request id=0x4dd7, seq=3/768, ttl=64 (reply in 26)  |
| 26  | 162.629756 | 192.168.1.1       | 192.168.1.10      | ICMP     | 98     | Echo (ping) reply id=0x4dd7, seq=3/768, ttl=64 (request in 25)  |
| 27  | 163.631018 | 192.168.1.10      | 192.168.1.1       | ICMP     | 98     | Echo (ping) request id=0x4ed7, seq=4/1024, ttl=64 (reply in 28) |
| 28  | 163.634334 | 192.168.1.1       | 192.168.1.10      | ICMP     | 98     | Echo (ping) reply id=0x4ed7, seq=4/1024, ttl=64 (request in 27) |
| 29  | 164.636556 | 192.168.1.1       | 192.168.1.10      | ICMP     | 98     | Echo (ping) request id=0x4fd7, seq=5/1280, ttl=64 (reply in 30) |
| 30  | 164.637652 | 192.168.1.1       | 192.168.1.10      | ICMP     | 98     | Echo (ping) reply id=0x4fd7, seq=5/1280, ttl=64 (request in 29) |
| 31  | 165.630092 | 0c:b2:f7:47:00:00 | Private_66:68:00  | ARP      | 60     | Who has 192.168.1.10? Tell 192.168.1.1                          |
| 32  | 165.631516 | Private_66:68:00  | 0c:b2:f7:47:00:00 | ARP      | 60     | 192.168.1.10 is at 00:50:79:66:68:00                            |

Character encoding: ASCII (0)  
[Coloring Rule Name: ICMP]  
[Coloring Rule String: icmp || icmpv6]  
Ethernet II, Src: Private\_66:68:00 (00:50:79:66:68:00), Dst: 0c:b2:f7:47:00:00 (0c:b2:f7:47:00:00)  
Destination: 0c:b2:f7:47:00:00 (0c:b2:f7:47:00:00)  
Source: Private\_66:68:00 (00:50:79:66:68:00)  
Type: IPv4 (0x0800)  
[Stream index: 5]  
Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.1  
0100 .... = Version: 4  
.... 0101 = Header Length: 20 bytes (5)  
Destination Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)  
Total Length: 84  
Identification: 0xd74e (55118)  
000. .... = Flags: 0x0  
...0 0000 0000 0000 = Fragment Offset: 0  
Time to Live: 64  
Protocol: ICMP (1)  
Header Checksum: 0xffff [validation disabled]  
[Header checksum status: Unverified]  
Source Address: 192.168.1.10  
Destination Address: 192.168.1.1  
[Stream index: 1]  
Internet Control Message Protocol  
Type: Echo (ping) request (8)  
Code: 0  
Checksum: 0xd130 [correct]  
[Checksum Status: Good]  
Identifier (BE): 20183 (0x4ed7)  
Identifier (LE): 55118 (0xd74e)  
Sequence Number (BE): 4 (0x0004)  
Sequence Number (LE): 1024 (0x0400)  
[Response Frame: 28]  
Data (56 bytes)  
Data: 0000000000000000f01112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f3031323334353637...  
[Length: 56]

0000 0c b2 f7 47 00 00 00 50 79 66 68 00 00 00 45 00 ...G...P yf  
0010 00 54 d7 4e 00 00 40 01 1f ff c0 a8 01 0a c0 a8 ...T.N...@...  
0020 01 01 08 00 d1 30 4e d7 00 04 00 09 0a 00 0c 0d ......0H...  
0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d .....  
0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d ...!"#\$%&'...  
0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d ..../012345 67  
0060 3e 3f .....>?

2026-09-20

Лабораторная работа №2

Гашимова Эсма Эльшан кызы

32 / 36

## 6. 6. Результаты работы

---

## 6.1 Выполненные действия

В GNS3 построена простая локальная сеть на базе Ethernet-коммутатора и двух VPCS.

Настроена IP-адресация конечных устройств и проверена связь между ними.

С помощью Wireshark проанализированы протоколы ARP и ICMP, а также работа проверки соединения в UDP- и TCP-режимах.

Построены и настроены сети с маршрутизаторами FRR и VyOS.

## 6.2 Полученные результаты

Для VPCS настроены адреса сети **192.168.1.0/24**.

Для маршрутизаторов FRR и VyOS настроен адрес **192.168.1.1/24**.

Работоспособность построенных топологий подтверждена успешным выполнением команды **ping**.

В Wireshark зафиксированы ARP-, ICMP-, UDP- и TCP-пакеты.

## 6.3 Вывод

В результате выполнения лабораторной работы получены практические навыки создания и настройки сетевых топологий в GNS3.

Освоена настройка IP-адресации VPCS, маршрутизаторов FRR и VyOS.

Изучены механизмы работы протокола ARP и передачи ICMP-, UDP- и TCP-пакетов.

Анализ трафика в Wireshark подтвердил корректную работу созданных сетей и настроенных устройств.